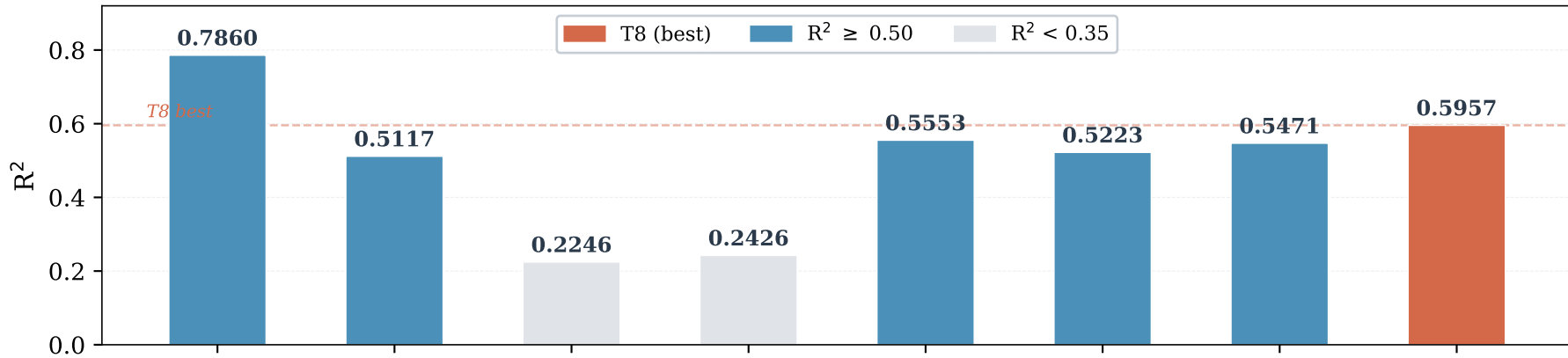
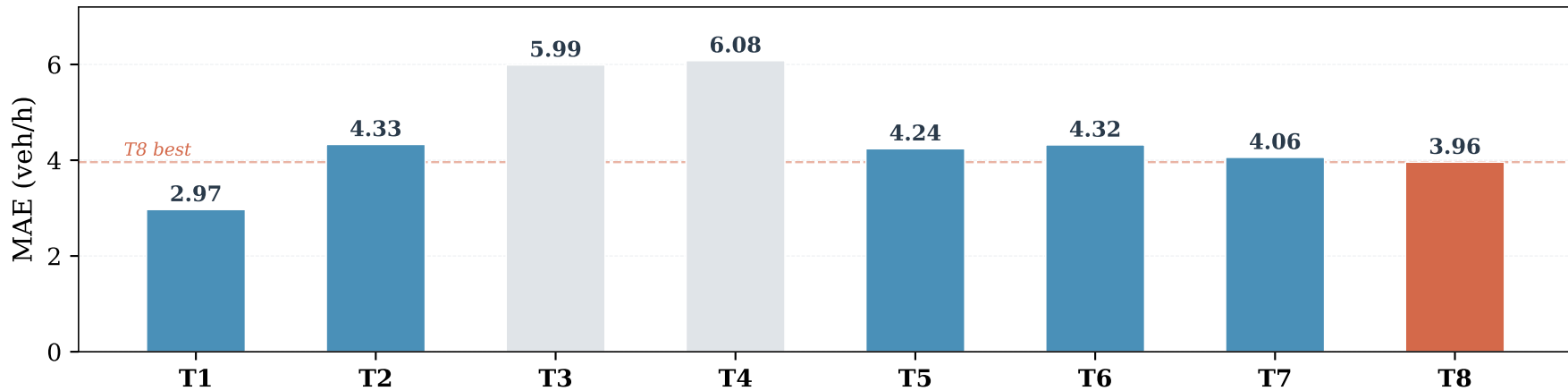


# Trial Progression T1→T8: Architecture Changes and Performance

(a)  $R^2$  per Trial (higher = better)



(b) MAE per Trial (lower = better)



|             | T1      | T2      | T3      | T4      | T5      | T6      | T7       | T8       |
|-------------|---------|---------|---------|---------|---------|---------|----------|----------|
| Final Layer | Linear† | GATConv | GATConv | GATConv | GATConv | GATConv | GATConv  | GATConv  |
| Dropout     | 0.0     | 0.3     | 0.0     | 0.0     | 0.3     | 0.3     | 0.3      | 0.2      |
| Batch Size  | 32      | 16      | 16      | 16      | 8       | 8       | 8        | 8        |
| LR          | 1e−3    | 5e−4    | 5e−4    | 5e−4    | 5e−4    | 3e−4    | 6e−4     | 5e−4     |
| Weighted    | No      | No      | wMSE    | wMSE    | No      | No      | No       | No       |
| Split       | 80/15/5 | 80/15/5 | 80/15/5 | 80/15/5 | 80/15/5 | 80/15/5 | 80/10/10 | 80/10/10 |

† T1: `use_dropout=False` → effective dropout = 0.0